Outlook

Re: Questions about the analysis

From Adam Wang <adam.wang@jcu.edu.au>
Date Tue 29-Oct-24 10:08 PM
To ian.hussey@unibe.ch <ian.hussey@unibe.ch>; Kaiqin Chan <Kaiqin.chan@jcu.edu.au>
Cc malte.elson@unibe.ch <malte.elson@unibe.ch>

Hi Ian

Thanks for the detailed email, we certainly appreciate the reminder and the word of caution. I have cc'd in the analyst for this project - Kaiqin, as he is probably in a better position to discuss stats-related aspects of this project. I do recall us having a protracted dicussion with the Editor regarding these issues such as fixed vs random effects, so these are certainly important issues.

@Kaiqin Chan are you able to respond to Ian's concerns?

Cheers
Adam

From: ian.hussey@unibe.ch <ian.hussey@unibe.ch>
Sent: Tuesday, October 29, 2024 7:30 PM
To: Adam Wang <adam.wang@jcu.edu.au>
Cc: malte.elson@unibe.ch <malte.elson@unibe.ch>
Subject: Questions about the analysis

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Hi Adam,

How are things going with data collection, what proportion of sites have returned their data?

I spent some time looking at the preregistered code for the analysis. I appreciate that folk often don't want to touch analysis plans once they have been approved at Stage 1, but at the same time my experience with RRRs is that issues in the analysis can come back to haunt you later when trying to determine what to conclude in your discussion and your results have a mixture of results. While we are still blind to the full results - or at least Malte and I are - we request that some of these details are ironed out.

I know these emails from me probably feel like a pain in your ass in a project that has, no doubt, taken very much time and effort for you. I want to again say that my motivation for saying all of this is to prevent the project stumbling at the last hurdle. I don't know if you followed it, but the very recent Protzko et al scandal (https://www.nature.com/articles/s41562-023-01749-9) has put an even brighter spotlight on meta-science and replication studies like this, and reminded us all of the pressure to get it right. The retraction of that article due to a poorly specified preregistration and poor adherence to it caused those authors a lot of stress and ultimately a retraction. I want to avoid the same thing happening here.

I haven't put together an exhaustive list here or done each of them in great detail, but I figured it's more important to start by putting this on your radar rather than delaying and missing our window for discussing this.

Currently the analyses seem to involved three, somewhat incompatible analyses:

An ANOVA, using data from all sites.
- This ignores the random effect for data collection site.
- This includes stimulus word as a fixed effect.

A meta-analysis of effects from an anova fit to each site's data
- This includes the random effect for data collection site.
- This ignores stimulus words as a fixed effect.

A multilevel model
- This ignores the random effect for data collection site (although comments mention also meta-analysing the three way interaction extracted from each site's multilevel model? See below)
- This ignores stimulus words as a fixed effect.

From my understand above, these three analyses therefore each take different positions on whether stimulus word and data collection site should be modelled and how (random vs fixed effects). Best practices for multilevel modelling, IMO, would say that both should be modelled as random effects, as both are non-exhaustive variables that increase the generalisability of the results.

Another issue is that the comments suggest that the three way interactions from multilevel models fit to each site's data will then be extracted and separately meta-analysed (i.e., in a different meta-analysis to the above mentioned one) - at least as far I can gather from the comments. This is unnecessarily complex: if all data were simply included in this model, and data collection site was included as a nested random effect, e.g., (1 | site/participant), this would achieve the same goal. Multilevel model followed by separate meta-analysis is fairly unusual as a modelling decision.

What is the estimand? I understand that as a replication study you may wish to provide a set of analyses that map on to the original study's analyses, hence the three way interaction. I could understand and support such an analysis in the service of fidelity of replication. At the same time, this article can provide the most highly powered test of these claims to date - but which claims specifically? The "ironic effect of thought suppression" mentioned in the title/abstract refers only to the suppression vs concentration conditions, and not to the other within/between factors or indeed the interaction effects. Currently, the mix of different analyses and the interpretation of the model parameters rather than predictions risk poor fit between verbal hypothesis/estimand and actual statistical test. Even if the direct replication of the original analyses are retained and labelled as such, I would strongly encourage that we provide a well specified model that provides a direct estimate of our desired estimand here. If I have missed something in how these all come together I am of course happy to be wrong, but I don't see it in the current manuscript or code.

More minor points are that comments in the code suggest that some factors and their associated interactions will be dropped from analyses if non-significant. This is generally a bad practice because a) it undermines the confirmatory nature of the analyses, as they become results dependent, and b) it makes the mistake of treating non-significant p values as evidence for the null hypothesis.

Lastly, and I don't have good documentation of this one at hand, but my recollection is that some of the analyses listed in the manuscript are at odds with the preregistered code, and that the preregistered code is incomplete. Given that updates to the code have to be made either way to a) finish them and b) align them with the preregister document, I think there is already scope to integrate the above issues with the currently planned analyses.

Best wishes
Ian

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